

SARAL RECRUITMENT TASK

GENERAL INSTRUCTIONS

- **Part A - Programming task** (build a working prototype + code)
- **Part B - Paper-reading & short presentation** (video ≤ 5 mins + slide summary)

You may rely on external literature, APIs, and prebuilt tools (LaTeX, Beamer, pandoc, PyLaTeX, LLMs, RAG, LangChain etc.). Cite every external resource. Submissions will be checked for plagiarism (including AI-generated content).

Feel free to write to us (reply-all in the email that you received with task instructions) for questions & clarifications.

SARAL Chatbot: Audience-Adaptive Script & Bullet Generator

Goal: Design a chatbot module for SARAL that, given a slide deck or summary, produces audience-adaptive **scripts, bullet points, and tweet-sized abstracts**. It should support iterative editing via conversation (user asks to “make it more visual”, “dumb down #3”, etc.).

Context & motivation:

Users should be able to chat with SARAL, drop in a slide or paper, and say “make a 90-second script for policymakers” or “create 10 tweet threads and 5 LinkedIn summaries” and iterate until satisfied.

See RAG literature for retrieval + generative setup. [arXiv+1](#)

Part A - Programming Task (required)

Implement a small **SARAL Chatbot** prototype that:

1. **Architecture requirements:** Use a retrieval layer (dense or sparse) over the uploaded paper(s) and a generator (LLM, e.g., open weights or API) to produce scripts and bullets. The pipeline should:
 - Ingest a paper (PDF/LaTeX) → embed chunks (DPR / sentence-BERT or OpenAI/other embeddings).

- Given a user prompt like: *"Make a 7-slide talk for graduate students focusing on methods; include 3 speaker notes per slide and preserve key equations"*, the system returns:
 - Slide-level bullets
 - Full speaker script (approx 5–7 minutes)
 - Speaker notes with math preserved in LaTeX inline
- Provide provenance: for each generated claim/sentence, link to the source chunk(s) or page numbers.

2. Core features:

- Produce multiple script lengths (30s, 90s, 5min) and styles (technical, plain-English, press release).
- Generate slide-level bullet points for each audience.
- Provide **change-tracking**: when the user asks "make #2 less technical," the chatbot must show the delta (old vs new) and why it changed.
- Scalable Architecture
- Support safety/style constraints (no offensive content, accessible language).

3. Implementation expectations:

- Implement RAG (retrieval-augmented generation) for long decks: chunk slides, create embeddings, and allow the agent to reference specific slides.
- Provide a minimal UI (chat interface) or a script showcasing the conversation flows.
- Include conversation logs as evidence that the bot can iterate and refine outputs
- Provide a clean architecture diagram

4. Evaluation:

- **Automatic**: factuality proxy by overlap between generated claims and source sentences (exact-match / semantic similarity); citation coverage (what % of content has a retrieved provenance).
- **Quality**: ROUGE / BERTScore vs. human-authored script for a small test set (3 papers).
- **Human eval**: audience appropriateness, factuality, and helpfulness (3 raters).

5. Implementation constraints: You may use existing embedding models: - open ai embeddings, google ai embeddings, (sentence-BERT) and small LLMs (e.g., 7B open models) or API-based LLMs. Use small conference papers to avoid excessive cost.

Deliverable: GitHub repo with a modular retrieval + generation pipeline, example runs (inputs → outputs), and evaluation.

Starter references: Classic RAG paper for design choices. [arXiv+1](#)

Part B - Paper-reading Task (required)

- Read the RAG NeurIPS 2020 paper and one recent slide-generation or summarization paper; produce a 5-minute video summarizing how RAG helps reduce hallucinations in this setting (connect to Part A's provenance and citation-coverage requirements), and propose one concrete improvement for SARAL's use-case - good candidates include math-aware retrieval (boosting chunks containing target equations), delta-anchored re-generation to keep change-tracked edits grounded, or an audience-conditioned re-ranker.
- Deliverable: 5-minute video + 1-page plan describing retrieval index construction, a chunking strategy that preserves LaTeX math blocks intact, and a prompt template family parameterised by {audience} (policymakers, grad students, press), {length} (30s / 90s / 5min), {style} (technical / plain-English / press release), and {change_instruction} for iterative edits - with at least two instantiated examples.
- **PS: Make a ppt/presentation based on your understanding for the interview, strictly it should be done in 15 minutes.**